



Agentic AI

Day 2: Use Cases, Agent Types & Future Trends

Learning Objectives

What you'll explore today



01

Understand real-world Use Cases of Agentic AI



02

Classify the 5 core Types of Agents



03

Analyze agent decision-making processes



04

Explore emerging Future Trends

Learning Outcomes

By end of Day 2, you will be able to...



Explain

Real-world AI agent applications across industries



Identify

Which agent type fits a given scenario



Apply

Agentic AI concepts to design your own agent



Evaluate

Trade-offs in autonomous AI systems

Recap: Day 1 Highlights

The Agentic AI Foundation



RAG AI

Retrieve → Augment → Generate
Static, one-shot responses

Agentic AI

Think → Plan → Act → Loop
Dynamic, autonomous, iterative

Agentic AI Use Cases

Where agents are changing the world



Healthcare

01



E-commerce

02



Coding

03



Business

04

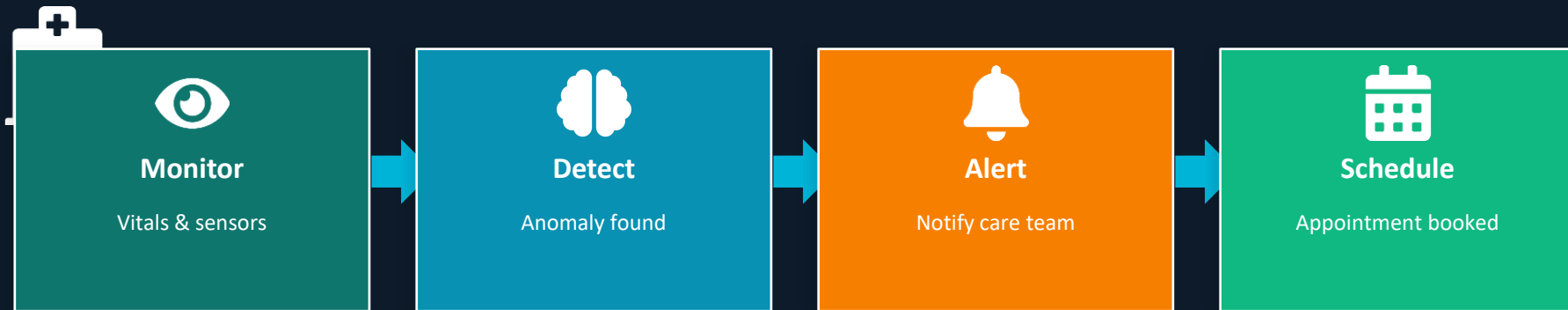


Autonomous
Vehicles

05

Healthcare AI Agent

Autonomous monitoring & response



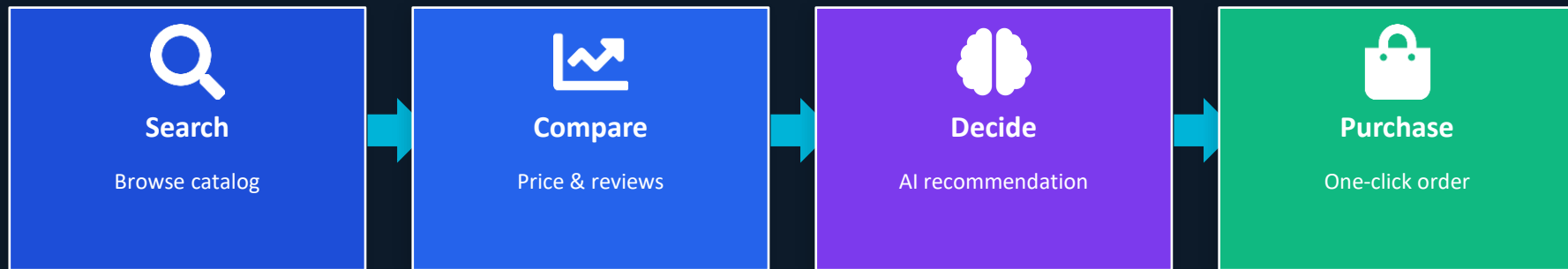
Key Benefit: 24/7 autonomous monitoring reduces response time by up to 60%



Used in ICUs, remote patient monitoring, chronic disease management

E-commerce AI Agent

Personalized shopping at scale

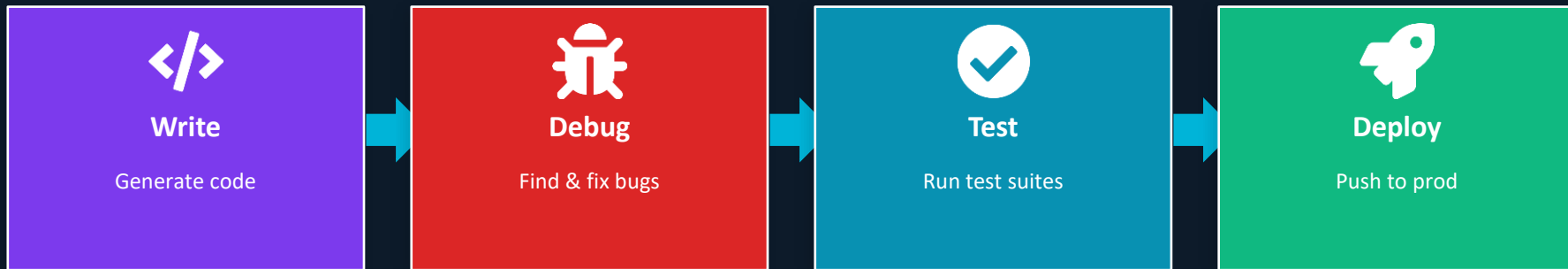


Impact: Personalized AI agents increase conversion rates by 35%+

 Amazon, Flipkart, Shopify all use agentic AI for shopping automation

Coding Agents

AI that writes, tests & ships code



Examples: GitHub Copilot, Devin, Cursor — autonomous coding assistants

 Developers using AI code agents ship 2× faster with fewer bugs

Business Automation

AI handling back-office operations



HR Workflows

Resume screening, onboarding, leave approvals



Invoice Processing

Extract, validate, reconcile, approve payments



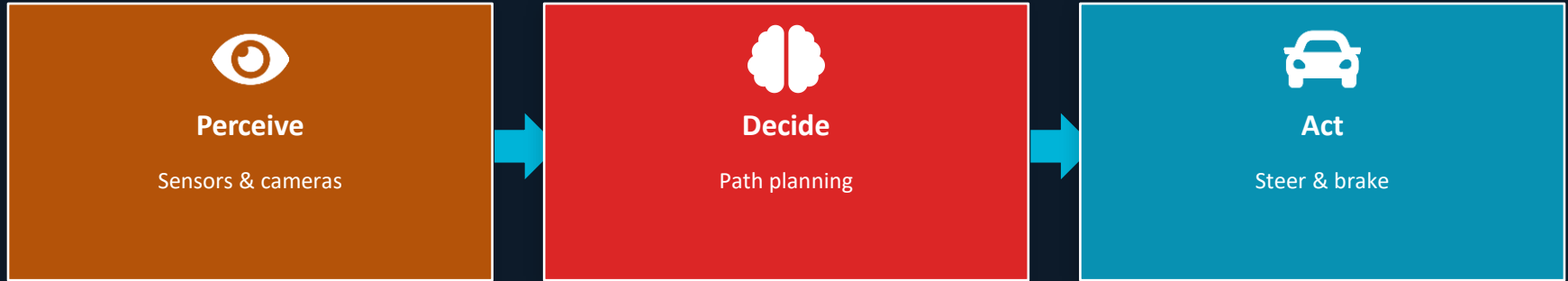
Analytics & Reports

Auto-generate KPI dashboards & insights

Business agents reduce manual workload by 40–70%, freeing humans for strategic tasks

Autonomous Vehicles

Real-time AI decision-making on the road



Key Tech

LiDAR • RADAR • Computer Vision
Deep RL • HD Maps

Examples

Tesla Autopilot • Waymo
Cruise • ISRO Autonomous Rovers



Quick Quiz!

Which of these is truly an Agentic AI task?

A Generate an image from a prompt

B Answer a single question

C Book a flight autonomously ✓

D Summarize a document



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Class Poll

Which AI use case is most impactful?



Healthcare

35%



E-commerce

25%



Autonomous
Vehicles

20%



Coding

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20%

5 Types of AI Agents

From simple reflexes to full learning systems



01

Reflex



02

Model-Based



03

Goal-Based



04

Utility-Based



05

Learning

Simple ————— Complex

Reflex & Model-Based Agents

Foundational agent architectures



Reflex Agent

- Rule-based decisions
- No memory or learning
- Condition $\rightarrow$ Action
- Example: Spam filter, Thermostat



Model-Based Agent

- Maintains internal memory
- Tracks world state over time
- Better at partial visibility
- Example: Navigation GPS agent

Goal-Based & Utility-Based Agents

Planning and optimizing outcomes



Goal-Based Agent

- Plans to achieve goals
- Considers future consequences
- Uses search & planning
- Example: Chess AI, Route Planner



Utility-Based Agent

- Optimizes for best outcome
- Uses utility/reward functions
- Handles trade-offs & uncertainty
- Example: Stock trading AI

Learning Agent

The most powerful — improves from experience



Learning Element

Improves agent from feedback & data

Performance Element

Takes actions using current knowledge

Critic

Evaluates performance vs standard

Problem Generator

Suggests new experiences to explore

Activity: Classify the Agent!

Which agent type is each?



Spam Filter

→ Reflex Agent



Google Maps

→ Model-Based Agent



Netflix AI

→ Utility-Based Agent



ChatGPT

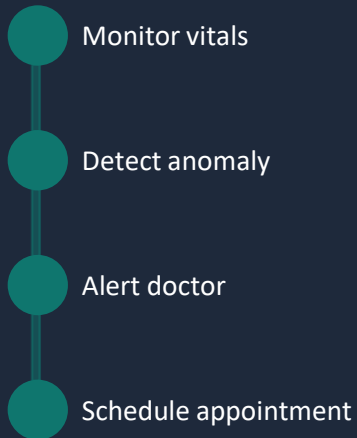
→ Learning Agent

Case Studies

Agents in real scenarios



Hospital AI



Smart Shopping



Self-Driving Car



Future Trends in Agentic AI

Where we're headed



Multi-Agent Systems

Multiple AIs collaborate on complex tasks



Autonomous Systems

End-to-end automation without human input



AI + IoT

Smart homes, factories & cities



Human-AI Teaming

AI as collaborative work partner

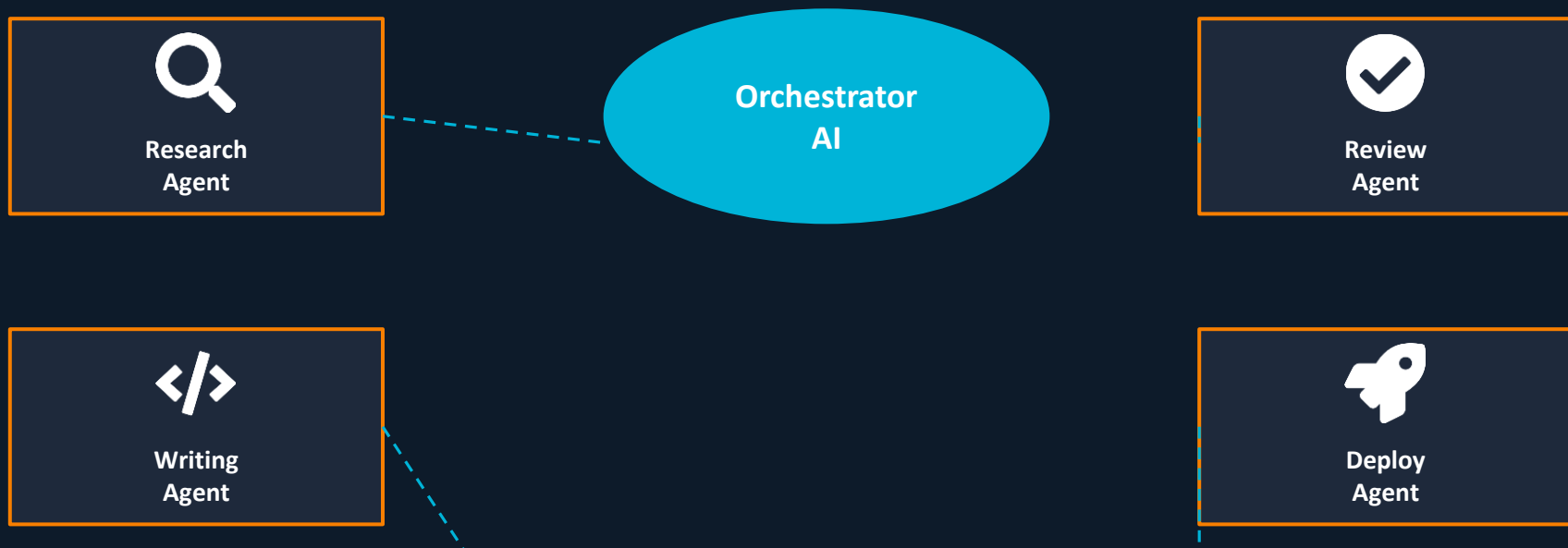


Ethical AI

Bias-free, safe & explainable systems

Multi-Agent Systems

Collaboration at scale



Use case: AI orchestrates research → writing → review → deployment autonomously

Ethical AI

Building responsible autonomous systems



Bias-Free

Train on diverse, representative datasets



Privacy

Data minimization, consent & encryption



Safety

Fail-safes, kill switches, human oversight

"AI should serve humanity — not replace human judgment in critical decisions"

Final Scenario

Which AI agent takes the wheel?

You say: 'Plan my trip to Goa & book tickets'



Answer: Goal-Based + Learning Agent

Autonomously plans, decides, acts — no human prompting needed!

Assignment

Design Your Own AI Agent



01 Choose a Domain

Healthcare, Education, Finance, Gaming...



02 Define the Goal

What problem does your agent solve?



03 Pick Agent Type

Reflex / Goal / Utility / Learning...



04 Draw the Flow

Perceive → Reason → Act → Learn loop



Thank You!

Day 2: Agentic AI — Use Cases, Types & Future

Questions & Discussion

Built with Agentic AI principles 🤖